

Task-Sufficient Contraction: Source Selection for Machine Information Interfaces

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Abstract

A declared task can sometimes certify a reduced source before a downstream encoder, codebook, rate, distortion target, or optimizer is chosen. This paper studies when one such reduction preserves the complete downstream problem family, a property termed Task-Sufficient Contraction. The reduced source is fixed by the task before the later operating point is selected. An exact contraction allows the later problem to be solved on that source with the same result as if the full source had been retained.

For a machine with a fixed set of possible actions and a fixed loss, the paper identifies a consumer-specific source by merging states only when every available action has the same regret in both. For finite action sets, replacing the richer source by this reduced source preserves the complete one-step rate-regret curve, even though the reduction is fixed before the distortion target is chosen. A second result gives an exact characterization for quadratic loss on affine feasible-action sets: the canonical reduced source is the projection onto the directions in which feasible actions can differ. Under a fixed energy budget, this becomes centered load, while retaining only the optimal water-filled action is too coarse. Earlier Information Bottleneck, semantic rate-distortion, and goal-oriented quantization results are then used to distinguish exact, architecture-conditioned, approximate, failed, and corrected contractions. The framework suggests a way for heterogeneous machines to exchange what a receiving task needs without first aligning their full internal representations.

1 Introduction

A weather forecast cannot be sufficient for the weather. It can describe the conditions likely to matter over a region and time interval, but it cannot settle every physical consequence of what will actually occur. It cannot determine whether one particular gust, interacting with the exact local mechanics of one particular building, will dislodge a roof tile. Yet consider a wind turbine controller making a high-wind cut-out decision. Real turbine systems make high-wind operating and shutdown decisions from wind-speed thresholds, and wind forecasts are used operationally to anticipate those cut-outs (Smith et al., 2012; Chou et al., 2025). For that declared decision, the forecast’s prediction of average wind speed over the relevant period is sufficient. The weather remains richer than the information problem presented to the machine.

What changed is not the weather and not the forecast. What changed is the question.

Task-aware compression and representation learning already ask which information matters to a downstream task. The claim here is more specific. Sometimes the task can identify a smaller source before the later information problem is optimized. That reduction is not merely the representation

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chosen for one rate, distortion target, codebook size, or other operating point. The same reduced source remains valid across the range of operating points covered by the result.

The central question is therefore:

Can a declared task determine a reduced source before the downstream encoder, codebook, rate, distortion target, or optimizer is chosen, such that solving the downstream problem after the reduction gives the same result as solving it on the richer source?

In the exact case the requirement is simple to state:

reduce first, then optimize = optimize on the richer source.

Such reductions are called *Task-Sufficient Contractions*.

The distinction becomes more consequential when machines increasingly consume one another’s outputs. In an agentic economy, a model, agent, router, controller, or service may need to act on another system without access to its internal representation. A certified task-sufficient interface can let the receiver act without first reconstructing or aligning the producer’s internal state, provided every distinction capable of changing the receiving decision is preserved. The value of that possibility grows with the number and heterogeneity of machine-to-machine handoffs. Opaque-agent protocols already make this interface boundary an explicit engineering concern (A2A Protocol Working Group, 2026).

The paper makes three connected contributions. First, it develops a common certification framework for task-determined source reductions that remain exact across a declared downstream optimization family, with the reduction fixed before the later operating point is chosen. Second, for a fixed consumer, it identifies the consumer-specific regret-complete source G_C and proves that, for finite action sets, replacing the richer source by G_C leaves the entire one-step rate-regret curve unchanged. The reduction is therefore fixed before the distortion target is selected. Third, for quadratic loss on an affine feasible-action set, the paper characterizes the exact source directly from the directions in which feasible actions can differ. Under fixed-energy scheduling this source is centered load, while the optimal water-filled action is too coarse.

The framework grew out of several earlier reductions. An Information Bottleneck result, *Source-Side Sufficiency for the Information Bottleneck: Exact Reduction and Finite-Block Equivalence*, was first posted in April 2026 (Armstrong, 2026c). There, a sufficient statistic can replace the source before any point on the information curve is selected. A separately developed semantic rate-distortion result then showed an exact reduced problem created by a designed semantic source and its induced loss (Armstrong, 2026d). The companion paper *Task-Relative Information Contracts: Sufficiency, Evidence, and Hierarchical Preservation in Multi-Agent Systems* calls the interface that preserves everything X tells a receiver about a named criterion Y a *task-relative information contract*, or TRI contract (Armstrong, 2026a). It also studies how such a contract can be generated, what level of detail finite evidence can support, and when the same parent-facing contract survives hierarchical refinement.

The earlier Information Bottleneck, semantic rate-distortion, and goal-oriented quantization results are revisited as evidence for the framework, not as three new contributions. Their comparison distinguishes exact pre-optimization contraction from architecture-conditioned reduction, approximate reduction, candidate failure, and corrected contraction. In that sense the retrospective sections explain why several independently developed reductions belonged to the same reduce-first problem.

The theory is static. It does not claim that a reduction licensed for the current problem remains valid after acting through the interface changes the state, learner, candidate set, action menu,

criterion, or dependence structure. It also does not compare task-sufficient sources across a sequence of changing decision problems or define the direction in which representation resolution should change when the action-consequence structure moves. Those recursive questions are deliberately left outside the present paper.

2 Contraction before optimization

Task-Sufficient Contraction is not one universal formula. The recurring structure is:

$$\boxed{\text{richer source} \longrightarrow \text{task-certified reduced source} \longrightarrow \text{later optimization.}}$$

The reduced source depends on the problem. In the Information Bottleneck it is a sufficient statistic $\phi(T)$. When every later use of a criterion Y must remain possible, it is the TRI contract S_Y . For the consumer-specific construction below it is $G_{\mathcal{C}}$. In constrained quadratic goal-oriented quantization it is the part of the load vector that can change one feasible action relative to another.

A useful certification pattern has four parts.

1. The reduction is fixed from the stated task before the later operating point is chosen.
2. Every feasible full-source solution has a reduced counterpart that preserves the task-relevant quantity and is no worse in the optimized resource.
3. Every feasible reduced solution can be pulled back to an admissible full-source solution with the same task value.
4. Any additional constraints claimed by the theorem survive the reduction.

The proof takes a different form in each problem. In the IB result, an encoder is averaged over source states that the relevance variable cannot distinguish, and a reduced solution can be used again on the full source. The rate-regret theorem below does the same thing for source states that give every action the same regret. In semantic rate-distortion, exactness depends on the chosen architecture and loss. In constrained GOQ, the right reduction is found by asking how feasible actions can actually differ.

Figure 1 also shows the relationship to the companion TRI-contract paper. The contract S_Y keeps everything X tells the receiver about Y . A particular consumer may need less. The two reductions can therefore be nested without making one a required first step for the other.

3 Consumer-specific task-sufficient sources

Let X denote the producer-side source and let Y be the criterion used to judge the receiver's output. A declared consumer specifies the actions it can take and the loss used to evaluate them. X is taken to have finite support in the main definitions so that the reduced source is finite. The rate-regret theorem later also assumes finitely many actions. The continuous GOQ example does not use that coding theorem and is handled directly.

Definition 1 (Machine consumer). A *machine consumer* is a pair

$$\mathcal{C} = (\mathcal{A}, \ell),$$

where $\mathcal{A}$ is the set of admissible outputs or actions and $\ell : \mathcal{A} \times \mathcal{Y} \rightarrow \mathbb{R}$ is the loss by which those outputs are evaluated.

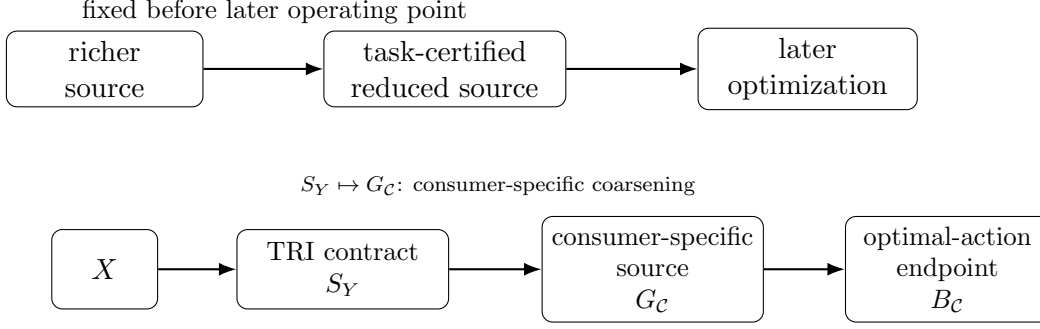


Figure 1: Top: the general Task-Sufficient Contraction pattern. Bottom: one important nested case, $X \rightarrow S_Y \rightarrow G_C \rightarrow B_C$. Here S_Y is the TRI contract and preserves everything X tells the receiver about the named criterion, G_C preserves the distinctions that can change the relative cost of the consumer’s available actions, and B_C keeps only the optimal full-information action. The lower arrows denote deterministic factorisation or coarsening, not causal flow. This lower chain is one application of TSC, not a required sequence for every problem.

The definition does not depend on how the machine is implemented. The consumer may be a rule set, classifier, optimizer, controller, language model, or compound agent. What matters here is only what it can output and how those outputs are judged.

For each source state x , define the conditional risk

$$r_x(a) := \mathbb{E}[\ell(a, Y) \mid X = x], \quad a \in \mathcal{A}, \quad (1)$$

and the full-information Bayes envelope

$$b(x) := \inf_{a \in \mathcal{A}} r_x(a). \quad (2)$$

Assume $b(x)$ is finite for every source state and define the conditional regret

$$g_x(a) := r_x(a) - b(x) \geq 0. \quad (3)$$

The quantity $g_x(a)$ is how much worse action a is than the best action at state x . Because these values depend on the conditional law of Y given X , the reduced source depends on both the source law and the declared consumer.

Producer-state task losses are included as a special case: for any $L(x, a)$, take $Y = X$ and $\ell(a, Y) = L(Y, a)$. This is the form used later in the GOQ scheduling example.

3.1 Optimal action and the regret-complete source

When the Bayes-optimal action is unique for every source state, write

$$a^*(x) := \arg \min_{a \in \mathcal{A}} r_x(a).$$

Definition 2 (Bayes quotient). Under unique Bayes actions, define

$$x \sim_B x' \iff a^*(x) = a^*(x').$$

A canonical representative is $B_C := a^*(X)$.

This fixed-loss object is the Bayes quotient of Sevetlidis (2026a) and coincides with policy equivalence in the exact single-cycle formulation of Walsh (2026b). It retains what is needed to implement the full-information optimum.

Knowing only the best action is not always enough to design an imperfect interface. Compression or quantization can make the receiver choose a non-optimal action, and the cost of those mistakes can differ even when the best action is the same.

Definition 3 (Regret-profile equivalence and consumer-specific source). Define

$$x \sim_G x' \iff g_x(a) = g_{x'}(a) \text{ for every } a \in \mathcal{A}. \quad (4)$$

Let

$$G_{\mathcal{C}} := [X]_{\sim_G}$$

be the resulting equivalence-class label. $G_{\mathcal{C}}$ is called the consumer-specific regret-complete source. The same object is the regret-profile quotient.

Two source states receive the same value of $G_{\mathcal{C}}$ when every available action is worse than the local optimum by the same amount in both states. The states may still differ in their absolute baseline loss. What has been removed is only information that cannot change the relative cost of any action available to this consumer.

Using regret as a distortion measure is already established in sequential decision-making and appears explicitly in Walsh’s single-cycle formulation (Russo and Van Roy, 2022; Arumugam and Van Roy, 2021; Walsh, 2026b). Here the whole set of action regrets is used for a different purpose: to decide which source states still need to be kept separate before the later problem is solved.

Definition 4 (Regret-complete statistic). A deterministic statistic $Q = q(X)$ is *regret-complete* for $(P, \mathcal{C})$ if there exists a function d_Q such that

$$g_x(a) = d_Q(q(x), a) \quad \text{for all } x \in \mathcal{X}, a \in \mathcal{A}.$$

Proposition 1 (Coarsest pointwise regret-complete quotient). *If $Q = q(X)$ is regret-complete, then there exists a deterministic map h with*

$$G_{\mathcal{C}} = h(Q).$$

Thus $G_{\mathcal{C}}$ is the coarsest deterministic quotient of X that preserves every action regret pointwise, up to relabelling.

This minimality is pointwise. It says that any deterministic summary that preserves the cost of every action must also distinguish every value of $G_{\mathcal{C}}$. It does not claim that $G_{\mathcal{C}}$ is the smallest summary that happens to preserve one particular rate-regret curve or one chosen distortion level.

Proof. If $q(x) = q(x')$, regret-completeness gives $g_x(a) = d_Q(q(x), a) = d_Q(q(x'), a) = g_{x'}(a)$ for every a . Hence $x \sim_G x'$, so each fibre of Q lies inside one regret-equivalence class. The class label is therefore a deterministic function of Q . $\square$

Proposition 2 (Regret quotient refines the Bayes quotient). *Under unique Bayes actions, there exists a deterministic map k such that*

$$B_{\mathcal{C}} = k(G_{\mathcal{C}}).$$

Proof. Within a regret-equivalence class the complete regret function is identical. Its unique zero is therefore identical, and that zero is the unique Bayes-optimal action. $\square$

In plain terms, B_C records what the consumer should do with full information. G_C also records what each possible mistake would cost.

Recent work on action-sufficient goal representations shows that information sufficient to recover the optimal value need not recover the optimal action distribution (Hyeon et al., 2026). For state S , goal G and representation $Z = \phi(S, G)$, their action-sufficiency condition is $I(A; G \mid S, Z) = 0$, where A follows a fixed optimal policy. That condition concerns optimal action prediction. The present construction asks a stronger preservation question for imperfect interfaces: which source distinctions preserve the relative cost of every feasible action? Under unique Bayes actions, B_C retains the optimal-action endpoint, whereas the consumer-specific regret-complete source G_C retains the complete feasible-action regret profile. Under the finite-source and finite-action assumptions of Theorem 1, the latter preserves the entire one-step rate-regret curve with the reduction fixed before the distortion target is selected. Two states can therefore share the same optimal action while remaining distinct because the same nonoptimal action incurs different regrets.

Proposition 3 (When the Bayes quotient is regret-complete). *Under unique Bayes actions, the Bayes quotient is regret-complete if and only if there exists a function*

$$d : \mathcal{A} \times \mathcal{A} \rightarrow \mathbb{R}_{\geq 0}$$

such that

$$g_x(a) = d(a^*(x), a) \quad \text{for every } x \text{ and } a. \quad (5)$$

Proof. If (5) holds, every regret profile is determined by $a^*(x)$. Conversely, if the regret profile is determined by the Bayes quotient, define $d(a^*, a)$ to be the common regret value on that Bayes class. $\square$

Equation (5) is exactly the condition under which sending only the optimal action also preserves the cost of every possible mistake.

3.2 Simple examples

Squared error. Let $Y \in \mathbb{R}$ and $\ell(a, Y) = (Y - a)^2$. Writing $m(x) = \mathbb{E}[Y \mid X = x]$,

$$r_x(a) = \text{Var}(Y \mid x) + (a - m(x))^2,$$

so $g_x(a) = (a - m(x))^2$. Bayes action and regret profile are both generated by $m(X)$.

Log loss. For finite Y , let the action be a predictive distribution q and $\ell(q, y) = -\log q(y)$. If $p_x = P(Y \mid X = x)$, then the Bayes action is p_x and

$$g_x(q) = D_{\text{KL}}(p_x \parallel q).$$

Again the Bayes and regret quotients coincide with posterior equivalence.

Zero-one classification. Let $Y \in \{1, \dots, K\}$ and actions be class labels. Then

$$g_x(a) = p_{\max}(x) - P(Y = a \mid x), \quad p_{\max}(x) = \max_j P(Y = j \mid x).$$

The Bayes quotient keeps only the maximizing class, but the complete regret vector determines the whole posterior. Indeed,

$$\sum_a g_x(a) = K p_{\max}(x) - 1,$$

so

$$p_{\max}(x) = \frac{1 + \sum_a g_x(a)}{K}, \quad P(Y = a \mid x) = p_{\max}(x) - g_x(a).$$

Thus two inputs with the same Bayes label can be distinct consumer-specific source states even under ordinary classification loss.

3.3 Relation to the full posterior about the criterion

Let

$$S_Y(X) := P(Y \in \cdot \mid X)$$

denote the posterior of Y given X . The companion TRI-contract paper (Armstrong, 2026a) uses S_Y for the interface that preserves everything X tells the receiver about Y . Since every action cost depends on X through this conditional distribution, every action-regret profile is also a function of S_Y .

Proposition 4 (Posterior, regret and Bayes hierarchy). *There exist deterministic maps h_C and k_C such that*

$$S_Y \longrightarrow G_C \longrightarrow B_C$$

under unique Bayes actions, where the arrows denote deterministic factorisation.

When the criterion and consumer are considered together, the three objects have a simple ordering. S_Y keeps everything X tells the receiver about Y . G_C keeps only what can change the relative cost of an action available to this consumer. B_C keeps only what is needed to identify the best action. This ordering connects the two papers. It does not mean that Task-Sufficient Contraction must always begin by constructing S_Y .

3.4 Loss decomposition

Suppose an information interface produces a representation U from X , and a receiving rule produces A from U , so $Y - X - U - A$ is a Markov chain. Then

$$\mathbb{E}[\ell(A, Y)] = \mathbb{E}[b(X)] + \mathbb{E}[g_X(A)]. \quad (6)$$

The first term is the loss that remains even with full information. The interface can affect only the second term, which measures the extra loss caused by choosing a non-optimal action. The next question is whether evaluating that extra loss really requires the full source X .

4 Task-Sufficient Contraction

Assume in this section that the action set $\mathcal{A}$ is finite. Following the established rate-regret formulation, define

$$R_X(D) := \inf_{\substack{p(a|x): \\ \mathbb{E}[g_X(A)] \leq D}} I(X, A). \quad (7)$$

By data processing, an intermediate representation can be replaced by the distribution over actions that it induces without increasing mutual information, as in the single-cycle formulation of Walsh (2026b). The source-side result in Armstrong (2026c) makes the complementary move: it averages an encoder over source states that a sufficient statistic already treats as identical. The theorem below asks the same reduce-first question for the consumer-specific source. The consumer is fixed

before the distortion target D is chosen. The question is whether the same reduced source works for the whole rate-regret curve.

Because $g_x(\cdot)$ is constant inside each G_C fibre, define

$$d_G(g, a) := g_x(a) \quad \text{for any } x \text{ with } G_C(x) = g, \quad (8)$$

which is well defined, and let

$$R_G(D) := \inf_{\substack{p(a|g): \\ \mathbb{E}[d_G(G_C, A)] \leq D}} I(G_C, A). \quad (9)$$

Theorem 1 (Rate-regret invariance under consumer-specific contraction). *For every achievable D ,*

$$\boxed{R_X(D) = R_G(D)}. \quad (10)$$

Proof. Take any feasible action channel $p(a | x)$ and average it within each regret-equivalence fibre:

$$\bar{p}(a | g) := P(A = a | G_C = g).$$

The averaged channel preserves the joint law of (G_C, A) and hence preserves expected regret, while data processing gives

$$I(G_C, A) \leq I(X, A).$$

Thus $R_G(D) \leq R_X(D)$. Conversely, any G_C -channel pulls back to X by $p(a | x) := p(a | G_C(x))$. Then $A - G_C - X$ and G_C is a function of X , so $I(X, A) = I(G_C, A)$ and the expected regret is identical. Hence $R_X(D) \leq R_G(D)$. $\square$

The averaging step is classical sufficiency and rate-distortion machinery (Bahadur, 1954; Witsenhausen, 1980). What matters here is when it is applied. The consumer-specific source is fixed before D is selected, yet no point on the one-step rate-regret curve improves by returning to the richer source.

Under unique Bayes actions, the zero-regret endpoint collapses further to the Bayes quotient. In the finite setting,

$$R_X(0) = H(B_C),$$

a result already explicit in the policy-equivalence rate-regret formulation of Walsh (2026b). The Bayes quotient therefore describes the zero-regret endpoint. Preserving the regret of every feasible action can require distinctions absent from the Bayes quotient. The source G_C preserves these regrets pointwise and is sufficient for the entire one-step rate-regret curve. This does not assert that it is minimal among all statistics that happen to preserve that optimized curve.

Theorem 1 is stated for finite $\mathcal{A}$. The continuous-action quadratic results below do not use this coding theorem. Instead they compare the full regret functions directly.

4.1 Five contraction regimes

In practice, a designer may propose a simpler task variable $Q = q(X)$ before knowing whether it really can replace the richer source. Five cases should be kept separate.

Exact consumer-complete contraction. If Q is regret-complete, equivalently if

$$g_X(a) = d_Q(Q, a) \quad \forall a,$$

then Proposition 1 gives $G_C = h(Q)$. The problem can contract at least as far as G_C . If Q is merely a relabelling or refinement of that quotient, solving on Q is exact but not minimal.

Architecture-conditioned contraction. For an arbitrary deterministic $Q = q(X)$, define the induced distortion

$$d_Q(q, a) := \mathbb{E}[g_X(a) \mid Q = q]. \quad (11)$$

If the architecture deliberately restricts the downstream action channel to factor through Q , so $X - Q - A$ is Markov, then

$$\mathbb{E}[g_X(A)] = \mathbb{E}[d_Q(Q, A)].$$

This includes stochastic encoders and intermediate representations generated from Q . The construction is classical indirect distortion (Witsenhausen, 1980). It gives the correct loss for a system that has chosen to work only through Q . It does not show that the consumer itself would be unable to use information that Q discarded.

Surrogate contraction. A proposed Q may instead support a simpler approximate loss $\tilde{d}(Q, a)$ satisfying

$$\alpha \tilde{d}(Q, a) \leq g_X(a) \leq \beta \tilde{d}(Q, a) \quad (12)$$

in a stated regime. Working on Q then comes with a bound, but it is not an exact reduction.

Candidate reduction failure. A proposed task variable may merge states that still assign different costs to some admissible action. The variable is then too coarse for exact contraction.

Corrected contraction. Failure of the obvious task variable need not mean that the full source is irreducible. A different reduced source may still preserve the downstream problem exactly. The constrained GOQ example below is the main instance.

These cases answer different questions. Does the reduced source preserve the whole later problem being claimed? Is it exact only because the architecture forces the system to use that source? Does it provide only an approximation? Or was the wrong reduced source chosen in the first place?

5 Neighbouring theories and attribution

Task-Sufficient Contraction sits beside several established ideas rather than replacing them. Classical sufficiency and Blackwell comparison ask when a reduced observation preserves decision value (Blackwell, 1953; Bahadur, 1954). Indirect rate-distortion gives the correct loss when the encoder observes a variable different from the one on which performance is judged (Witsenhausen, 1980). Functional compression asks which source distinctions must survive when the receiver needs a function of the source rather than a reconstruction of the source itself (Orlitsky and Roche, 2001).

Bruhacs (2026) formulates task-relative preservation under projection and the least quotient sufficient for a declared task family. Those results provide a close formal comparison for task-relative reduction. The present paper develops a certification framework for stated downstream optimization families, together with the consumer-specific regret construction and affine-action quadratic characterization. The general idea that a task determines which distinctions must survive is not claimed as new.

Recent work develops consumer-specific reductions. Sevetlidis (2026a) studies Bayes-sufficient representations for a fixed loss. Sevetlidis also studies loss shift, in which changing the loss can make a previously sufficient representation inadequate even when the joint data distribution remains fixed (Sevetlidis, 2026b). This concerns transfer between declared losses. The rate-regret result here fixes the consumer and asks whether one source reduction remains exact across the stated distortion targets. Russo and Van Roy (2022) and Arumugam and Van Roy (2021) connect information

acquisition and near-optimality through rate-distortion ideas. Most directly, Walsh (2026b) uses policy regret as distortion and identifies policy equivalence as the exact zero-regret endpoint of a single-cycle rate-regret problem. These results establish important pieces of the decision-aware picture.

Hyeon et al. (2026, Appendix I) also study Q-sufficiency, which preserves the optimal Q-function, and derive action sufficiency under a specified tie-breaking rule. The regret-complete source here preserves relative action costs while discarding action-independent loss baselines.

Wei et al. (2026) distinguish mechanism, response, and decision sufficiency for world-model planning. For a fixed query and candidate set, their response object retains absolute candidate costs, while the decision object retains a selected optimum. With the candidate set and cost criterion matched, the absolute cost profile determines the regret profile used here, which in turn determines the optimum under this paper’s unique-optimum assumption. Their planner analysis also identifies normalized relative-cost weights as sufficient for MPPI updates and explicitly notes common-offset invariance. Thus relative-cost preservation is not absent from their analysis. Their work distinguishes the information required by particular planner operations, including search and final selection. Theorem 1 here certifies preservation of the complete one-step rate-regret curve by one fixed regret-complete source.

The common question developed here is whether a task-certified reduction can be fixed before the later rate, codebook, distortion target, or other operating choice, while preserving the stated downstream problem family. The framework distinguishes exact consumer-complete contraction, architecture-conditioned contraction, bounded approximation, candidate failure, and corrected contraction. The constructions in this paper show how those different guarantees can be established and why they should not be conflated.

Related equivalence ideas appear elsewhere. Myhill–Nerode minimisation merges automaton histories that no permitted continuation can distinguish with respect to language acceptance (Nerode, 1958). Observational equivalence separates internal identity from external behaviour in process theory (Hennessy and Milner, 1985). Bisimulation-based state abstraction preserves controlled behaviour over transitions (Ferns et al., 2004). These are useful analogies. Their dynamic closure requirements are not part of the static theory developed here.

6 Three earlier reductions

The three reductions in this section are earlier results, not three new contributions of the present paper. Their comparison exposed the common reduce-first pattern. Each case is therefore presented in two parts: what the earlier paper already established, and what the comparison adds here.

6.1 Information Bottleneck: exact pre-optimization contraction

Native result. Let T be the source and C the relevance variable. Consider the probabilistic prediction consumer with action space $\Delta(\mathcal{C})$ and logarithmic loss. For

$$p_t = P(C \mid T = t),$$

the Bayes action is p_t , and the regret of predicting q is

$$g_t(q) = D_{\text{KL}}(p_t \parallel q).$$

Under log loss, two source states are therefore equivalent exactly when they induce the same posterior over C .

The existing source-side sufficiency theorem assumes a deterministic $\phi(T)$ satisfying

$$C \perp T \mid \phi(T).$$

Thus the conditional distribution of C is determined by $\phi(T)$, so $\phi(T)$ contains everything needed to compute the log-loss regret of any prediction.

The companion IB result proves that every full encoder has a fibre-averaged shadow on $\phi(T)$ with the same relevance and no greater source rate, and that

$$I(U, T) = I(U, \phi(T)) + I(U, T \mid \phi(T)). \quad (13)$$

The complete IB curve, the Lagrangian infima, the minimisers and the finite-block log-loss problem are therefore unchanged by solving first on $\phi(T)$ (Armstrong, 2026c).

In the present terminology, IB is the clean exact case:

criterion-certified quotient $\longrightarrow$ exact optimization contraction.

TSC reading. The native theorem says more than that a good compressed representation exists. It says that the IB optimization never needed all distinctions in T in the first place. This makes IB the cleanest exact example of contraction before a later operating point is chosen. The theorem itself remains the result of the earlier paper.

6.2 Semantic rate-distortion: architecture-conditioned contraction

Native result. The semantic/source-coding example starts from a producer state T and a designer-chosen deterministic oracle allocation

$$Q = \phi^*(T).$$

The underlying consumer is evaluated by welfare loss $\ell(t, a)$.

Under general smooth concave utility, knowing the deterministic oracle $\phi^*(T)$ is not enough by itself to recover the cost of every other action. The semantic rate-distortion paper therefore defines the reduced loss as

$$d_s^*(q, a) := \mathbb{E}[\ell(T, a) \mid \phi^*(T) = q]. \quad (14)$$

Any reconstruction rule on the full source can then be averaged over states with the same $Q = \phi^*(T)$. This leaves the reduced loss unchanged and cannot increase mutual information. Conversely, any rule defined on Q can be used on T through the map $T \mapsto Q$. The semantic marginal rate-distortion problem can therefore be solved exactly on $\phi^*(T)$ with the induced loss d_s^* (Witsenhausen, 1980; Armstrong, 2026d).

Here d_s^* is the expected task loss after conditioning on the reduced source, matching the semantic rate-distortion formulation. Written in the regret notation used here, the corresponding loss is

$$d_Q^{\text{reg}}(q, a) = d_s^*(q, a) - \mathbb{E}[b(T) \mid \phi^*(T) = q].$$

The subtracted term does not depend on the chosen action once q is fixed. It therefore changes the baseline loss but not which action is better. The raw-loss and regret versions describe the same reduced decision problem.

For $\phi^*(T)$ itself to preserve the cost of every possible mistake, additional structure is needed. It holds, for example, when

$$\ell(t, a) - \ell(t, \phi^*(t))$$

is a function only of $(\phi^*(t), a)$, including the exact quadratic subclasses considered in the companion work.

The same example exposes a second issue. One interface may have to support both allocation and verification. If verification can be expressed as an ordinary per-item distortion, the two requirements still fit the same semantic rate-distortion problem. If verification depends on the occupancy of the whole pool, it no longer has that form, and the problem becomes a constrained feasibility problem instead (Armstrong, 2026d).

Thus the semantic example separates two statements:

a quotient can define an exact restricted problem

without implying

the chosen variable is the smallest exact source for the unrestricted consumer.

What the comparison adds. Comparing this case with IB exposes an important distinction. Information can be absent because the consumer has no use for it, or because the system design has chosen not to expose it. Those are not the same claim. The additional verification requirement shows the same point from another direction: a reduction that is sufficient for one declared requirement need not be sufficient for another.

6.3 Goal-oriented quantization: failure of the obvious source and exact correction

Native result. The current GOQ manuscript, *Task-Oriented Quantization for Quadratic Scheduling: Centroid Water-Filling and Power-Diagram Encoders*, supplies the constrained scheduling case (Armstrong, 2026b). Let T be the producer state and $\phi^*(T)$ the deterministic full-information optimum.

Exact interior case. For isotropic quadratic task loss,

$$\ell(t, a) = \frac{\gamma}{2} \|a - \phi^*(t)\|^2,$$

the entire regret profile factors through $\phi^*(t)$. The Bayes and regret quotients therefore coincide with the oracle action, and the task RDF and finite- K quantization problem reduce exactly to ordinary MSE coding on $\phi^*(T)$.

Smooth interior case. Under a zero-gradient interior optimum and smooth/strong curvature,

$$\frac{\alpha}{2} \|\phi^*(t) - a\|^2 \leq \ell(t, a) - \ell(t, \phi^*(t)) \leq \frac{\beta}{2} \|\phi^*(t) - a\|^2.$$

Distance between oracle actions then gives a controlled approximation to the true task loss. It yields the corresponding rate-distortion bounds and approximation guarantee, but it does not make the oracle action an exact reduced source.

What the comparison adds. The exact and smooth interior results make the oracle action look like a natural reduced source. The constrained case shows why choosing that source is a real problem rather than a change of terminology. Two source states can have the same optimal action while making other feasible actions cost different amounts. The answer is not to keep the whole raw source. It is to find the smaller source that still preserves those differences.

Quadratic losses on affine action sets. The correction yields a more general source rule. Let the feasible action set $\mathcal{A} \subseteq \mathbb{R}^n$ have affine hull

$$\text{aff}(\mathcal{A}) = a_0 + V,$$

where V is the linear subspace spanned by feasible action differences, and assume $\mathcal{A}$ is nonempty and that $\inf_{b \in \mathcal{A}} f(t, b)$ is finite for every source state under consideration, with

$$f(t, a) = \|t + a\|^2, \quad g_t(a) = f(t, a) - \inf_{b \in \mathcal{A}} f(t, b).$$

Theorem 2 (Affine-action quadratic regret quotient). *For any two source states t, t' ,*

$$g_t(\cdot) = g_{t'}(\cdot) \iff t' - t \in V^\perp.$$

Consequently the orthogonal projection

$$\boxed{\Pi_V t}$$

is a canonical representative of the regret-profile quotient.

Proof. Let $d = t' - t$. For any feasible a ,

$$f(t', a) - f(t, a) = \|t'\|^2 - \|t\|^2 + 2d^\top a.$$

If $d \in V^\perp$, then $d^\top a$ is constant on the affine set $a_0 + V$, so every feasible action value shifts by the same amount and the regret profile is unchanged. Conversely, if the regret profiles agree, then $f(t', a) - f(t, a)$ is constant over $a \in \mathcal{A}$. Hence $d^\top(a - a') = 0$ for every feasible a, a' . These differences span V , so $d \in V^\perp$. $\square$

The theorem has a simple interpretation. A component of the source that is orthogonal to every direction in which feasible actions can differ may change the absolute objective value, but it changes every feasible action by the same amount. It cannot change the cost of one action relative to another, so the consumer does not need it.

Fixed-energy scheduling. Fix a positive energy budget $E > 0$. For load states $t \in \mathbb{R}_{\geq 0}^N$ and

$$\mathcal{A}_E = \{a \in \mathbb{R}_{\geq 0}^N : \mathbf{1}^\top a = E\},$$

the affine direction is $V = \mathbf{1}^\perp$. Theorem 2 therefore yields

$$\boxed{G_C(t) = Pt = t - \bar{t}\mathbf{1}, \quad P = I - \frac{1}{N}\mathbf{1}\mathbf{1}^\top.} \tag{15}$$

This is centered load.

The Bayes-optimal action is the water-filling solution

$$\phi^*(t) = \text{WF}_E(t) = (\lambda(t)\mathbf{1} - t)^+,$$

but the oracle action is strictly too coarse once non-optimal actions are possible. For example, with $E = 1$,

$$t = (0, 10), \quad t' = (0, 20),$$

both have $\phi^*(t) = \phi^*(t') = (1, 0)$, yet for the feasible action $a = (0, 1)$,

$$g_t(a) = 20, \quad g_{t'}(a) = 40.$$

Thus identical Bayes actions do not imply identical regret profiles.

At the same time, the raw load contains a common shift that cannot change the relative cost of feasible actions. The exact GOQ assignment rule for fixed representatives therefore depends only on centered load because

$$2t^\top a_k + \|a_k\|^2 = 2(Pt)^\top a_k + \|a_k\|^2 + 2\bar{t}E,$$

and the last term is common to every k . The cell update can also be computed from centered load alone. If

$$\mu_k = \mathbb{E}[T \mid T \in \mathcal{C}_k],$$

then

$$\mu_k = \mathbb{E}[PT \mid \mathcal{C}_k] + \mathbb{E}[\bar{T} \mid \mathcal{C}_k]\mathbf{1},$$

and water-filling is invariant to common shifts, so

$$\text{WF}_E(\mu_k) = \text{WF}_E(\mathbb{E}[PT \mid \mathcal{C}_k]).$$

Hence both assignment and update are well defined on PT .

On the natural load state space, the constrained example therefore has the form

$$\boxed{T \text{ contains a null common mode} \quad \longrightarrow \quad T - \bar{T}\mathbf{1} \text{ is exact} \quad \longrightarrow \quad \phi^*(T) \text{ can be too coarse.}}$$

Failure of the obvious task representation does not imply that the producer source is irreducible. It can mean that the task induces a different source.

	IB	Semantic/RD	GOQ
Producer source	T	demand/state T	task/load state T
Declared consumer	predictor of C	allocation/semantic consumer	action consumer
Reduction status	exact reduction	architecture-conditioned generally	exact, approximate, or corrected reduction
Key object	sufficient $\phi(T)$	designed $\phi^*(T)$ with induced loss	oracle action or projected load
Boundary	insufficient statistic loses relevance	second contract / architecture restriction	active constraints change the right reduced source

Table 1: Three separately developed reductions viewed through the Task-Sufficient Contraction framework.

7 Machine information interfaces

The same idea has a direct systems interpretation. A producer may keep a rich internal representation H_P , while the receiving machine uses a different internal representation H_C . An interface between them does not necessarily need to expose either internal representation.

One approach is representation alignment. A sender and receiver may exchange embeddings, hidden states, or caches and learn a transformation that makes the sender's internal state usable by

the receiver. Recent latent-communication work makes this approach increasingly explicit, including cross-architecture alignment as an open design problem (Liu, 2026). This can be valuable when the receiving system needs broad access to the sender’s latent content.

Task-Sufficient Contraction suggests another possibility. If the receiver has a declared task, an external interface can be built around a certified reduced source rather than either party’s complete internal representation. Two producers with different internals may then be able to serve the same receiver without first making those internals mutually intelligible.

This distinction is particularly natural for agent protocols in which the remote system is intentionally opaque. A2A exposes capabilities and exchanged artifacts while allowing internal state, memory, tools, and proprietary logic to remain hidden (A2A Protocol Working Group, 2026). In that setting the practical question is not how to reproduce the remote agent internally. It is what information must cross the interface for the receiving task.

There need not be one universal interface for every receiver. A broad task may still require a rich representation, while a narrow task may require much less. One way to describe the alternative is *consequence alignment*: producers need not represent the world in the same way if they expose the distinctions that can change what the receiver should do.

This does not make internal representations irrelevant, and it does not imply that the reduced interface is cheap to compute. A rich latent representation may still be the best internal way to produce the information the receiver needs.

The systems implication is narrower:

Representation alignment can be sufficient for interoperability, but it need not be necessary when the receiving task admits a shared task-sufficient interface.

That proposition is testable. A later systems study can compare a representation-alignment pipeline with an interface constructed directly from a verified reduced source, measuring task performance, transmitted bits or tokens, latency, compute, memory and energy. The present paper stops before making those empirical claims.

8 Scope and boundaries

Every reduction in this paper has a stated scope. Exactness never means that the discarded information is useless for every possible future task.

If one interface must serve consumers

$$\mathcal{C}_1, \dots, \mathcal{C}_m,$$

with regret sources

$$G_1, \dots, G_m,$$

then the joint source

$$G_{1:m} := (G_1, \dots, G_m)$$

preserves the regret of every available action for all of them.

Proposition 5 (Static multi-consumer refinement). *Adding a declared consumer can only refine the joint regret source:*

$$G_{1:m} = h(G_{1:m+1})$$

for the projection h . In finite systems,

$$H(G_{1:m}) \leq H(G_{1:m+1}).$$

This is a static result. An interface that must serve several declared tasks may need to keep distinctions that any one of those tasks could ignore. The present paper does not claim that such an interface can anticipate tasks that were never part of its specification.

Several other boundaries matter.

Hidden or misspecified criteria. The reduction is only as good as the consumer specification. If the declared actions or loss leave out a consequence that actually matters, the reduced source may discard information the real problem needs.

Architecture versus consumer. A designer may intentionally force a system to work only through a reduced variable Q , even when the consumer could use more information. The loss in (11) then describes that restricted system correctly. It does not show that the consumer itself had no use for the discarded information.

Approximate reduction. A simpler reduced variable may give useful bounds without preserving exact regret. Smooth GOQ is an example. A bounded approximation should not be described as exact sufficiency.

Implementation cost. Information-theoretic redundancy is not the same as implementation cost. Removing a distinction from the external task does not prove that the cheapest implementation should avoid computing or storing it internally.

State-changing action and the recursive boundary. The present theory is static. Bisimulation-based state abstraction requires equivalence to survive state transitions (Ferns et al., 2004). Subramanian et al. (2022) separately require sufficiency for expected-reward evaluation and for predicting the compressed state’s own evolution in their information-state formulation. Walsh (2026a) studies recurrent consequence-sensitive compression, including policies that alter later inferential conditions. The present paper does not impose or solve these recursive requirements. If acting through an interface changes the learner, state distribution, criterion, candidate set, action menu, or dependence structure, a reduction valid for the current problem may no longer be valid for the next decision. The results here do not compare source requirements across such changes or define a dynamic update rule for the task-sufficient source.

9 Discussion

The weather example makes sufficiency concrete. A forecast that cannot settle whether a gust will dislodge a roof tile can still provide everything needed for a particular wind-turbine decision. Sufficiency belongs to the declared use of the information, not to the richness of the thing being described.

This becomes especially important when the consumer is another machine. An agentic economy creates repeated handoffs among models, agents, controllers, and services that may be opaque to one another. If a receiving machine can be given a certified task-sufficient interface, interoperability need not require sharing or aligning complete internal representations at every handoff. The systems value of a valid reduction therefore grows as machine-to-machine coordination becomes more common and heterogeneous.

Task-Sufficient Contraction says that the reduction can sometimes be made before the later optimization begins. First identify the source distinctions the task can never use. Then solve the

later information problem on what remains. An exact result means that returning to the richer source cannot improve any of the operating points covered by the theorem.

The three earlier reductions show why the framework needs more than one label. The Information Bottleneck provides the cleanest exact case. A sufficient source statistic can be fixed before the rate or Lagrange multiplier is selected, and the complete optimization family is unchanged. Designed-source semantic rate-distortion shows that an exact reduced problem can instead be created by an architectural restriction and an induced loss. Goal-oriented quantization shows that an obvious task representation can fail without making the raw source irreducible. Under fixed-energy quadratic scheduling, centered load is exact even though the optimal water-filled action is too coarse.

The consumer-specific regret construction gives another exact case. It keeps every distinction that can change the relative cost of an available action and discards distinctions that cannot. If the same problem is also described by a criterion Y , the relationship is:

$$X \longrightarrow S_Y \longrightarrow G_C \longrightarrow B_C.$$

The chain separates three questions. What information does X provide about Y ? Which of those distinctions can change the cost of an action available to this consumer? Which distinctions are still needed if only the best full-information action matters?

For machine interfaces, the implication is that shared internal representations are not always necessary. A producer may keep a rich internal state while exposing only a task-sufficient external object. Another producer with a different internal representation may expose the same external distinctions. This does not imply that every task admits a small interface or that a smaller interface is cheaper to compute. It says only that the external information requirement should be derived from the receiving problem rather than assumed to be the producer’s full representation.

The static boundary matters. A reduction that is valid now need not remain valid after the receiver acts. Action may change the learner, state distribution, candidate pool, action menu, criterion, or dependence structure. The next question is therefore not another static contraction at one fixed state. It is how the information required by the task changes across states, and whether the representation should become finer or coarser as the action-consequence problem changes. This paper does not answer that recursive question.

References

- A2A Protocol Working Group. Agent2agent (a2a) protocol specification. Version 1.0.0, 2026. <https://a2a-protocol.org/v1.0.0/specification>.
- Joss Armstrong. Task-relative information contracts: Sufficiency, evidence, and hierarchical preservation in multi-agent systems. Zenodo preprint, 2026a. All versions DOI: 10.5281/zenodo.22819849.
- Joss Armstrong. Task-oriented quantization for quadratic scheduling: Centroid water-filling and power-diagram encoders. Zenodo preprint, 2026b. doi: 10.5281/zenodo.22819867.
- Joss Armstrong. Source-side sufficiency for the information bottleneck: Exact reduction and finite-block equivalence. arXiv preprint arXiv:2604.26744v2, 2026c.
- Joss Armstrong. Designed-source reductions and a dual-purpose feasibility band for semantic rate-distortion. arXiv preprint arXiv:2606.11280, 2026d. URL <https://arxiv.org/abs/2606.11280>.
- Dilip Arumugam and Benjamin Van Roy. Deciding what to learn: A rate-distortion approach. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings*

- of *Machine Learning Research*, pages 373–382. PMLR, 2021. URL <https://proceedings.mlr.press/v139/arumugam21a.html>.
- R. R. Bahadur. Sufficiency and statistical decision functions. *The Annals of Mathematical Statistics*, 25(3):423–462, 1954. doi: 10.1214/aoms/1177728715.
- David Blackwell. Equivalent comparisons of experiments. *The Annals of Mathematical Statistics*, 24(2):265–272, 1953. doi: 10.1214/aoms/1177729032.
- Lorand Bruhacs. Task-sufficient inference and explanation: Reasoning under projection, 2026. URL <https://philarchive.org/rec/BRUTIA-9>. Draft preprint, 19 March 2026.
- Shih-Chun Chou, Hui-Ling Chang, Zoltan Toth, Pay-Liam Lin, Jing-Shan Hong, Yuan-Kang Wu, Guan-Ru Chen, Kuo-Chen Lu, and Chia-Ping Cheng. Evaluation, calibration, and application of probabilistic 100-m wind speed forecasts produced by the wrf ensemble prediction system in taiwan. *Journal of Applied Meteorology and Climatology*, 64(1):3–19, 2025. doi: 10.1175/JAMC-D-23-0210.1.
- Norm Ferns, Prakash Panangaden, and Doina Precup. Metrics for finite markov decision processes. In *Proceedings of the Conference on Uncertainty in Artificial Intelligence*, pages 162–169, 2004.
- Matthew Hennessy and Robin Milner. Algebraic laws for nondeterminism and concurrency. *Journal of the ACM*, 32(1):137–161, 1985. doi: 10.1145/2455.2460.
- Jinu Hyeon, Woobin Park, Hongjoon Ahn, and Taesup Moon. Action-sufficient goal representations. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pages 49233–49262. PMLR, 2026. URL <https://proceedings.mlr.press/v306/hyeon26a.html>.
- Yingzhuo Liu. Beyond tokens: a unified framework for latent communication in llm-based multi-agent systems. arXiv preprint arXiv:2606.05711, 2026.
- Anil Nerode. Linear automaton transformations. *Proceedings of the American Mathematical Society*, 9(4):541–544, 1958. doi: 10.2307/2033204.
- Alon Orlitsky and James R. Roche. Coding for computing. *IEEE Transactions on Information Theory*, 47(3):903–917, 2001.
- Daniel Russo and Benjamin Van Roy. Satisficing in time-sensitive bandit learning. *Mathematics of Operations Research*, 47(4):2815–2839, 2022. doi: 10.1287/moor.2021.1229.
- Vasileios Sevetlidis. Bayes-sufficient representations in supervised learning. arXiv preprint arXiv:2606.04045, 2026a.
- Vasileios Sevetlidis. Loss-shift transfer via Bayes quotients. arXiv preprint arXiv:2606.13178, 2026b.
- Joe Smith, Arlinda Huskey, Dave Jager, and Jerry Hur. Wind turbine generator system safety and function test report for the integrity ew50 wind turbine. Technical Report NREL/TP-5000-56587, National Renewable Energy Laboratory, November 2012. URL <https://www.nrel.gov/docs/fy13osti/56587.pdf>.

- Jayakumar Subramanian, Amit Sinha, Raihan Seraj, and Aditya Mahajan. Approximate information state for approximate planning and reinforcement learning in partially observed systems. *Journal of Machine Learning Research*, 23(12):1–83, 2022. URL <https://jmlr.org/papers/v23/20-1165.html>.
- Mark Walsh. Support sufficiency as consequence-sensitive compression in belief arbitration. arXiv preprint arXiv:2604.16434, 2026a.
- Mark Walsh. Support sufficiency as action-sufficient compression: A single-cycle rate-regret formulation. arXiv preprint arXiv:2606.09858, 2026b.
- Rongzhe Wei, Hans Hao-Hsun Hsu, Peizhi Niu, Yifan Li, and Pan Li. What must a world model distinguish for planning? arXiv preprint arXiv:2609.33030, 2026. Version 1, 26 September 2026.
- Hans S. Witsenhausen. Indirect rate distortion problems. *IEEE Transactions on Information Theory*, 26(5):518–521, 1980. doi: 10.1109/TIT.1980.1056251.